



DATABASE ENGINEERING

Complete Learning Roadmap

Design scalable database systems with strong data correctness, performance, and maintainability.

Relational modeling

Normalization

SQL query writing

Index basics

Transactions and isolation

Query optimization

01

Fundamentals

02

Core Concepts

03

Tools and
Frameworks

04

Real Projects

05

Advanced Topics

5

Learning Stages

20+

Core Topics

10+

Projects

100%

Database



Learning Path Overview

Five progressive stages from fundamentals to production mastery

01

Fundamentals

Relational modeling •
Normalization • SQL query
writing • Index basics

02

Core Concepts

Transactions and isolation •
Query optimization • Schema
migration planning • Backup
strategies

03

Tools and Frameworks

PostgreSQL/MySQL internals

04

Real Projects

High-traffic schema redesign •
Read replica rollout • ETL
pipeline integration • Audit
logging pipeline

05

Advanced Topics

Sharding approaches •
Disaster recovery architecture
• Data governance framework
• Capacity and cost planning

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.

Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- Relational modeling
- Normalization
- SQL query writing
- Index basics

Tools & Frameworks

- Relational modeling
- Normalization
- SQL query writing
- Index basics

Recommended Projects

- Database Engineering workshop using Relational modeling
- Database Engineering portfolio project with Normalization

Stage 1 Milestone Complete Database Engineering workshop using Relational modeling and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- Transactions and isolation
- Query optimization
- Schema migration planning
- Backup strategies

Tools & Frameworks

- Transactions and isolation
- Query optimization
- Schema migration planning
- Backup strategies

Recommended Projects

- Database Engineering workshop using Transactions and isolation
- Database Engineering portfolio project with Query optimization

Stage 2 Milestone Complete Database Engineering workshop using Transactions and isolation and document the decisions, trade-offs, and validation checks before moving to Stage 3.

Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- PostgreSQL/MySQL internals
- NoSQL trade-offs
- Replication tooling
- Monitoring tooling

Tools & Frameworks

- PostgreSQL/MySQL internals

Recommended Projects

- Database Engineering workshop using PostgreSQL/MySQL internals
- Database Engineering portfolio project with NoSQL trade-offs

Stage 3 Milestone Complete Database Engineering workshop using PostgreSQL/MySQL internals and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- High-traffic schema redesign
- Read replica rollout
- ETL pipeline integration
- Audit logging pipeline

Tools & Frameworks

- High-traffic schema redesign
- Read replica rollout
- ETL pipeline integration
- Audit logging pipeline

Recommended Projects

- Database Engineering workshop using High-traffic schema redesign
- Database Engineering portfolio project with Read replica rollout

Stage 4 Milestone Complete Database Engineering workshop using High-traffic schema redesign and document the decisions, trade-offs, and validation checks before moving to Stage 5.

Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

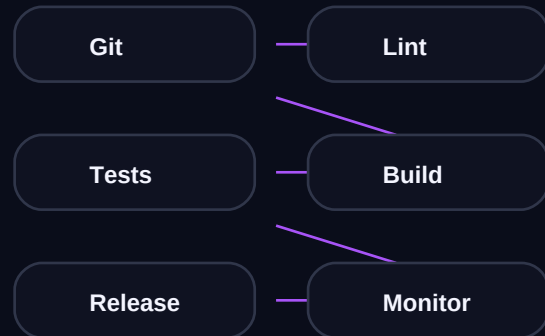
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- Sharding approaches
- Disaster recovery architecture
- Data governance framework
- Capacity and cost planning

CI/CD Pipeline



Stage 5 Milestone

A production-grade database engineering workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE

Database Engineering docs and community guides

Link: <https://www.postgresql.org/docs/>

FREE

Build one portfolio project per stage before moving ahead

Link: <https://roadmap.sh/postgresql-dba>

PAID

Structured specialization track focused on Database Engineering

Link: <https://roadmap.sh/postgresql-dba>

FREE

Database Engineering roadmap reference

Link: <https://roadmap.sh/postgresql-dba>

FREE

Official documentation

Link: <https://www.postgresql.org/docs/>

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in Database Engineering?
- Which project best proves Stage 4 competency in Database Engineering?
- What trade-offs appear in Stage 5 decisions for Database Engineering?